# CURRICULUM VITAE


## Thyagarajulu Gollapalli

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**Research Interests**

Numerical modelling; megathrust earthquakes; subduction dynamics; mantle convection; structural geology–geodynamics integration; artificial intelligence applied to geophysical datasets and earthquake processes.

**Education**

**Ph.D**, Apr. 2018–Nov. 2022
Computational Geodynamics & Geophysics, Monash University & Indian Institute of Technology (IIT) Bombay
Thesis: Unravelling tectonic coupling and loading along Sunda margin through 3-D regional numerical modelling
Advisors: Fabio A. Capitanio, Monash University & M. Radhakrishna, IIT Bombay

**Bachelor of Science**, Aug. 2011–May 2015
Indian Institute of Science (IISc), Bengaluru, India
Major: Mathematics
Minor: Earth & Environmental Science

**Professional Experience**

**Research Officer** — Jan. 2026–Present
Australian National University, Canberra, ACT

**Assistant Lecturer** — Jun. 2025–Dec. 2025
Monash University, Melbourne, VIC

**Research Officer** — Aug. 2023–Jun. 2025
Monash University and AuScope, Melbourne, VIC

**Imaging Geophysicist** — Nov. 2022–Jun. 2023
CGG, Perth, WA

**Junior Research Fellow** — May 2017–Apr. 2018
Gravity & Magnetics Lab, Department of Earth Sciences, IIT Bombay, Mumbai, India

**Project Assistant** — Aug. 2015–Mar. 2017
Computational Geodynamics Lab, Centre for Earth Sciences, IISc, Bengaluru, India

**Classes Taught**

EAE3032 (Undergraduate), Geophysical Interpretations in geoscience, geography and environmental science, Unit Coordinator, Monash University

EAE5053 (Postgraduate), Advanced Spatial Analysis and Modelling, Unit Coordinator, Monash University

**Teaching Assistant**

Physics of the Solid Earth, EAE3532, 2019

Earth, Atmosphere and Environment 1, EAE1011, 2020

**Fellowships Awards, and Achievements**

Indian Institute of Technology Joint Entrance Examination (IIT-JEE), 2011, ranked in top 1% of Indian applicants for undergraduate study

INSPIRE Scholarship for Higher Education, 2011–2015, Department of Science and Technology (DST), India

Graduate Aptitude Test in Engineering (GATE), 2016, ranked in top 20% in Geology & Geophysics

Junior Research Fellow Scholarship, 2017–2018, Ministry of Human Resource Development (MHRD), India

Faculty of Science Dean's International Postgraduate Research Scholarship, AUD $176,297, 2018–2022, Monash University

Monash Graduate Scholarship, AUD $27,353 per annum, 2018–2021, Monash University

AGU Fall Meeting Student Travel Grant, USD $1,000, 13–17 Dec. 2021

Graduate Research Completion Award (GRCA), approximately AUD $5,000, Dec. 2021–Jan. 2022, Monash University

Postgraduate Publications Award (PPA), AUD $5,151, Apr.–Jun. 2022, Monash University

**Scientific Softwares**

Underworld geodynamics code (developer), CitcomS, Litmod-3D, ASPECT, GPlates

**Skills**

**Scientific Computing:** Python, Matlab, C, Fortran, Julia, R, Shell scripting, AWK

**Data Analysis and Visualisation:** Matplotlib, Cartopy, Plotly, Seaborn, Scikit-learn, PyTorch

**Geospatial and Geophysical Tools:** GMT, QGIS, Surfer, ParaView, PyVista, GPlates

**Research Computing and Development:** Git, Docker, PyCharm, HPC workflows, reproducible workflows

**Core Competencies:** Data analysis, data visualisation, machine learning, deep learning, GIS, problem solving, leadership, adaptability, relationship building

**Instruments:** Gravimeter (CG-5 Autograv), Magnetometer (GST-19M Proton Precession), Isotope Ratio Mass Spectrometer (IRMS MAT 253), thin section and petrology lab instruments

**Internships**

**Undergraduate Research Assistant**, Thin Section Lab, Centre for Earth Sciences, IISc, Bengaluru, India, May 2014–Jul. 2014
**Research:** Delineating metamorphism effect in rock minerals along chilled margins
**Advisor:** Kusala Rajendran

**Undergraduate Research Assistant**, Mass Spectrometry Lab, Centre for Earth Sciences, IISc, Bengaluru, India, May 2013–Jul. 2013
**Research:** Latitudinal dependency of carbonates from dental enamel
**Advisor:** Prosenjit Ghosh

**Undergraduate Summer Intern**, Department of Mathematics, IISc, Bengaluru, India, May 2012–Jul. 2012
**Research:** Group theory reading and discussion sessions
**Advisor:** Kaushal Verma

# Publications

**In preparation**

**1.** **Gollapalli, T.**, Moresi, L., How to represent faults in geodynamic models.

**2.** Knight, B. S., Moresi, L., **Gollapalli, T.**, Graciosa, J. C., Lu, N., Discretisation errors in underworld3 for a range of benchmarks.

**3.** Khalil, H., **Gollapalli, T.**, Capitanio, F. A., Knight, B., & Brune, S., How far-field rotational motion drives rift evolution.

**Journal articles**

**5.** Capitanio, F. A., **Gollapalli, T.**, M. Radhakrishna., Graciosa, J. C., Zuhair, M., Beall, A., & Dal Zilio, L., Bridging the gap between subduction dynamics and the long-term strength of megathrusts, *Nature Communications*, doi:10.1038/s41467-025-65824-7, 2025.

**4.** Moresi, L., Mansour, J., Giordani, J., Knepley, M., Knight, B., Graciosa, J. C., **Gollapalli, T.**, Lu, N., & Beucher, R., Underworld3: Mathematically self-describing modelling in Python for desktop, HPC and cloud, *Journal of Open Source Software*, doi:10.21105/joss.07831, 2025.

**3.** Graciosa, J. C., Capitanio, F. A., Beall, A., Hargreaves, M., **Gollapalli, T.**, Tang, T., & Zuhair, M., Testing driving mechanisms of megathrust seismicity with explainable artificial intelligence, *JGR: Solid Earth*, doi:10.1029/2024JB028774, 2025.

**2.** Zuhair, M., **Gollapalli, T.**, Capitanio, F. A., Betts, P. G., & Graciosa, J. C., The role of slab step on tectonic loading along subduction zones: inferences on the seismotectonics of the Sunda convergent margin, *Tectonics*, doi:10.1029/2022TC007242, 2022.

**1.** Ghosh, A., **Gollapalli, T.**, & Steinberger, B., The importance of upper mantle heterogeneity in generating the Indian Ocean geoid low, *Geophysical Research Letters*, doi:10.1002/2017GL075392, 2017.

# Conference Posters & Presentations (selected)

**3.** **European Geosciences Union General Assembly**, oral session, 23–27 May 2022, “Megathrust seismicity through the lens of explainable artificial intelligence.”

2. **American Geophysical Union Fall Meeting**, oral session, 13–17 Dec. 2021, "Subduction dynamics and tectonic coupling along seismically active margin: the relevance of trench-parallel forces."

1. **American Geophysical Union Fall Meeting**, poster session, 9–13 Dec. 2019, "Megathrust seismotectonics of the Southeast Asian convergent margin: insights from three-dimensional spherical geodynamic modelling."

## Workshops

7. Underworld workshop at Curtin University, 2025.

6. Underworld workshop at AuScope HQ, 2024.

5. Underworld workshop at University of Sydney, 2023.

4. The Victorian Universities Symposium on Computational Geodynamics, 3 Nov. 2021, organised by University of Melbourne.

3. Graduate Research Conference, 19 Nov. 2019, organised by Monash University.

2. Quantitative Seismology and Microseismicity using Geophysical and Geochemical Techniques, 2017, organised by CEaS, IISc and Shell Technology Centre, Bengaluru.

1. Vijyoshi Camp, annual national science camp, 26–27 Nov. 2011, organised by IISc, Bengaluru.

## Blogs

4. Deep Earth forces reveal where the next megaquake may strike in Southeast Asia, 2025 [link].

3. Unlocking the secrets of great earthquakes with deep learning, 2022 [link].

2. Advanced geodynamic models of giant earthquakes, 2019 [link].

1. The missing mass—what is causing a geoid low in the Indian Ocean?, 2017 [link].

## Outreach & Service

**Open Day Demonstrator**, Monash, 2022, and IISc, 2016–2017.

**Protolith Volunteer**, biennial festival, IIT Bombay, 2017.

**Member of EAE Outreach Team**, Monash University, 2019.

## Interests

**Marathon running, cycling, badminton.**

1. Nike Melbourne Marathon, 2021, 42.2 km, 03:57:26
2. Wings for Life World Run, 2022, 18 km, 01:43:11
3. Great Ocean Road Marathon, 2024, 44 km, 04:03:32.